UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 001 · Introduction to second and higher order DE with constant coefficients · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·12	Unit 2 deck 001, slide 12 (rendered .pptx)
T001 00:17:38	Unit 2 AV-summary transcript 001 ("Introduction to higher order differential equations, complementary function and particular integral"), segment start
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s3 3d	Past paper: UE23MA141A Jan 2024 set 3, question 3.d

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1 Linear differential equations and their order

Definitions

Linear differential equation (slide). A differential equation in which the dependent variable and its derivative occur in first degree and are not multiplied together is a Linear Differential Equation. S001·6 · T001 00:05:13

Linear differential equation (lecture form). A differential equation is linear if (1) the coefficients of y and its derivatives are functions of x or constants (never of y), and (2) the degree of y and of each derivative is 1. T001 00:06:27 · T001 00:06:41

Non-linear differential equation. A differential equation which is not linear, e.g. $y^2\frac{dy}{dx} + \sin y = x^2e^{2y}$. S001·5 · T001 00:03:29

Order. The order of the highest-order derivative present in the differential equation. T001 00:05:51

Higher-order LDE. A linear differential equation of order 2 or more: $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = X$, where $a_1, a_2, \ldots, a_n$ are functions of x or constants and X is a function of x alone. S001·8 · T001 00:09:46

Statements in the class material

> *"A linear differential equation of first order is of the form $\frac{dy}{dx} + P \cdot y = Q$ where P and Q are functions of x alone."* S001·7

> *"A linear differential equation of order two is of the form $\frac{d^2 y}{dx^2} + P_1 \frac{dy}{dx} + P_2 y = X$ where P_1 and P_2 and X are functions of x alone."* S001·7

> *"A linear differential equation of order n is of the form $\frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + P_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + P_n y = X$ where $P_1, P_2, \ldots, P_n$ and X are functions of x alone."* S001·7

Key points

1. Linear means: y and its derivatives only to the first degree, never multiplied together, and coefficients free of y. $y\frac{dy}{dx}$, y^2, $\sin y$, e^y make an equation non-linear. ESA ★ S001·5 · S001·6 · T001 00:05:18 · T001 00:05:34
2. The order is the highest derivative present, not the highest power. ESA ★ T001 00:05:51 · T001 00:11:19
3. Unit 2 treats only linear equations of order 2 and higher; first-order and non-linear equations were the previous unit. ESA ★ T001 00:04:04 · T001 00:04:22

Exam traps

Exam trap. The lecture says "P_0 equal to 0" for the second-order form (T001 00:07:34); it must be $P_0 \neq 0$, otherwise the equation is not of order 2. T001 00:07:34 · T001 00:08:55

2 Constant or variable coefficients; homogeneous or not

Definitions

LDE with constant coefficients. In this type, the coefficient of derivatives are constants. S001·8 · T001 00:10:28

LDE with variable coefficients. In this type, the coefficient of derivatives $a_1, a_2, \ldots, a_n$ are functions of x. S001·9 · T001 00:11:33

Homogeneous LDE. RHS X is zero. S001·10 · T001 00:13:22

Non-homogeneous LDE. RHS X is a nonzero function of x. S001·10 · T001 00:13:34

Associated homogeneous equation. The equation obtained from a non-homogeneous LDE by replacing X by 0: for $f(D)y = X$ it is $f(D)y = 0$. T001 00:14:44 · T001 00:14:52

Key points

1. Classify before solving: constant coefficients (classes 002–006, operator methods) or variable coefficients (class 011: Legendre and Cauchy forms, 8 ESA occurrences). Homogeneous (CF only) or non-homogeneous (CF + PI). ESA ★★ S001·3 · S001·8 · S001·9 · ESA Jan-24 s3 3d
2. A constant term or a function of x on the right makes the equation non-homogeneous; only $X \equiv 0$ is homogeneous. ESA ★★ S001·10 · T001 00:13:34
3. Every non-homogeneous LDE has an associated homogeneous equation: the same left side with X replaced by 0. ESA ★★ T001 00:14:27 · T001 00:14:52

Exam traps

Exam trap. $2x^3y''' - (5x^2 + 2)y'' + xy' + y = x^2 \sin x$ is linear (variable coefficients). Functions of x in the coefficients do not make an equation non-linear; only y in a coefficient does. S001·9 · T001 00:12:08

3 The differential operator D and $f(D)$

Definitions

Differential operator D. $D = \frac{d}{dx}$, $D^2 = \frac{d^2}{dx^2}$, $D^3 = \frac{d^3}{dx^3}$, ..., $D^n = \frac{d^n}{dx^n}$. S001·11 · T001 00:15:16

Linear differential operator $f(D)$. $f(D) = D^n + a_1 D^{n-1} + a_2 D^{n-2} + \cdots + a_n$, so that the LDE reads $f(D)y = X$. S001·11 · T001 00:16:29

Statements in the class material

> *"Thus, in terms of operator notation* $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = X$ *can be written as* $D^n y + a_1 D^{n-1} y + a_2 D^{n-2} y + \cdots + a_n y = X$, $f(D)y = X$, *where* $f(D) = D^n + a_1 D^{n-1} + a_2 D^{n-2} + \cdots + a_n$ *called linear differential operator."* S001·11

Key points

1. Constant-coefficient ESA questions are written as $f(D)y = X$, e.g. $(D-2)^2 y = 8(e^{2x} + \sin 2x)$ and $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$; read D as $\frac{d}{dx}$. ESA ★★★ S001·11 · ESA Dec-23 3a · ESA Jan-24 s2 2d
2. $f(D)$ with constant coefficients behaves like an algebraic polynomial: factorise it into linear factors, in any order. ESA ★★ T001 00:16:49 · T001 00:17:07

Why?

Why is the operator notation useful?
With constant coefficients $f(D)$ is a polynomial: it can be factorised, and (class 002) $f(D)e^{mx} = f(m)e^{mx}$ turns the DE into an algebraic equation. Classes 003–006 also invert it, $y_p = \frac{1}{f(D)} X$.
ADDED FOR CLARITY S001·11 · T001 00:16:49

Exam traps

Exam trap. $f(D)$ commutes and factorises like a polynomial only with constant coefficients: $D(xDy) = xD^2 y + Dy \neq xD(Dy)$. ADDED FOR CLARITY T001 00:16:49

Exam trap. $D^2 y$ is $\frac{d^2 y}{dx^2}$, not $\left(\frac{dy}{dx}\right)^2$; and the last coefficient a_n multiplies y, so $(D^2 + 4)y = y'' + 4y$. ADDED FOR CLARITY S001·11

4 Linear independence and the complementary function

Definitions

Linearly dependent functions. A set of functions $f_1(x), f_2(x), \ldots, f_n(x)$ are said to be linearly dependent on an interval I if there exists constants $c_1, c_2, \ldots, c_n$ not all zero, such that $c_1 f_1 + c_2 f_2 + \cdots + c_n f_n = 0$. S001·13 · T001 00:19:05

Linearly independent functions. If not linearly dependent, the functions are called linearly independent: $c_1 f_1 + c_2 f_2 + \cdots + c_n f_n = 0 \Rightarrow c_1 = c_2 = \cdots = c_n = 0$. S001·13 · T001 00:19:59

Complementary function (CF). The complete solution $y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$ of the homogeneous LDE of order n, where $y_1, \ldots, y_n$ are n linearly independent solutions. S001·12 · S001·16 · T001 00:25:38

Statements in the class material

> *"Two functions y_1 and y_2 are linearly independent when $c_1 y_1 + c_2 y_2 \neq 0$ for non zero constants c_1 and c_2, i.e. $c_1 y_1 \neq -c_2 y_2 \;\rightarrow\; y_1 \neq \left(\frac{-c_2}{c_1}\right) y_2$."* S001·14

> *"If y_1 and y_2 are two linearly independent solutions of a second order homogeneous linear differential equation, then $c_1 y_1 + c_2 y_2$ is also a solution of the differential equation. It is the complete solution of the differential equation."* S001·15

> *"If the nth order LDE with constant coefficients $\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = 0$ has n linearly independent solutions $y_1, y_2, y_3, \ldots, y_n$, then their linear combination $c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$ is also a solution of the differential equation. $y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$, the complete solution is called COMPLEMENTARY FUNCTION of the differential equation."* S001·16

Key points

1. Two functions are independent when neither is a constant multiple of the other: e^x, e^{2x} and $\sin x, \cos x$ are; $e^x, 3e^x$ are not. ESA ★★ S001·14 · T001 00:22:00
2. Any linear combination of solutions of a homogeneous LDE is again a solution (superposition). With n independent solutions it is the complete solution, the CF. ESA ★★★ S001·15 · S001·16 · T001 00:24:11
3. "Very important concept": linearly independent and dependent functions (T001 00:18:47). Independence guarantees the n constants are genuinely different. ESA ★★ T001 00:18:47 · S001·13

Why?

Why must the solutions in the CF be linearly independent?
If $y_2 = k y_1$, then $c_1 y_1 + c_2 y_2 = (c_1 + k c_2) y_1$ has only one effective constant, so it cannot be the complete solution of a second-order equation, which needs two. Independence makes the n constants genuinely different.
S001·14 · S001·15 · T001 00:22:18

Exam traps

Exam trap. The lecture says $c_1y_1 + c_2y_2$ is a solution "only when y_1 and y_2 are linearly independent" (T001 00:23:40). Wrong: every combination of solutions of a homogeneous LDE is a solution. Independence is what makes it the *complete* solution. ADDED FOR CLARITY T001 00:23:40 · S001·15

Exam trap. Slide 001·14's "$c_1y_1 + c_2y_2 \neq 0$ for non zero constants" must be read as: the combination is not identically zero unless $c_1 = c_2 = 0$. Equivalently y_1/y_2 is not a constant. ADDED FOR CLARITY S001·14 · T001 00:22:00

Exam trap. Superposition holds for the homogeneous equation only. Two PIs of $f(D)y = X$ do not add to a PI: their sum solves $f(D)y = 2X$. ADDED FOR CLARITY S001·17

5 The general solution: $y = \text{CF} + \text{PI}$

Definitions

Particular integral (PI). A particular solution y_p of the non-homogeneous equation $f(D)y = X$: one solution, containing no arbitrary constant. S001·17 · T001 00:27:16 · T001 00:27:52

Complete (general) solution. $y = y_c + y_p$, CF plus PI. S001·17 · T001 00:27:34

Statements in the class material

> *"The general solution of the non homogeneous linear differential equation has two components: Complementary Function and particular Integral. The complementary function is the complete solution of homogeneous linear differential equation."* S001·12

> *"Consider a non homogeneous linear differential equation $f(D)y = X$. If $y = y_c$ is the complete solution of $f(D)y = 0$ and $y = y_p$ is a particular solution of the differential equation $f(D)y = X$, then the complete solution is $y = y_c + y_p$. The particular solution y_p is called PARTICULAR INTEGRAL."* S001·17

Key points

1. Every Unit 2 ESA question (19 distinct, 24 occurrences in 8 papers) is a non-homogeneous linear DE whose answer is $y = \text{CF} + \text{PI}$. Write both parts and add them; an answer with only the CF or only the PI is incomplete. ESA ★★★ S001·17 · ESA Dec-23 3a · ESA Jul-23 3b · ESA Jan-24 s2 2d
2. The CF holds all n arbitrary constants of an n-th order equation; the PI has none. Count the constants as a check. ESA ★★★ S001·16 · S001·17 · T001 00:25:38
3. The CF is the complete solution of the associated homogeneous equation $f(D)y = 0$ (put $X = 0$); find it first. ESA ★★★ S001·12 · T001 00:17:58 · T001 00:26:57

Why?

Why is $y_c + y_p$ the complete solution of $f(D)y = X$?
$f(D)$ is linear: $f(D)(y_c + y_p) = 0 + X$. Conversely, if y is any solution, $f(D)(y - y_p) = 0$, so $y - y_p$ is a solution of the homogeneous equation and is contained in y_c. Hence every solution has the form $y_c + y_p$.

ADDED FOR CLARITY S001·17

Why is the PI called "particular"?
It is one particular solution, with no arbitrary constants. Any other choice differs from it by a solution of $f(D)y = 0$, which the CF absorbs: $\frac{1}{2}e^{3x}$ and $\frac{1}{2}e^{3x} + e^{x}$ are both PIs of $y'' - 3y' + 2y = e^{3x}$ (W5.1).

ADDED FOR CLARITY S001·17 · T001 00:27:34

Exam traps

Exam trap. Put the constants in the CF only; the PI has none. Do not fit initial conditions to the CF alone: apply them to $y = y_c + y_p$. ADDED FOR CLARITY S001·17

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. No question targets this class directly (definitions, classification, linear independence and the Wronskian were never asked on their own: 0 of 8 papers). But every Unit 2 ESA question (19 distinct questions, 24 occurrences) is a non-homogeneous linear DE whose answer is $y = \mathrm{CF} + \mathrm{PI}$, and every constant-coefficient one is written in D-notation $f(D)y = X$. So the structure $y = \mathrm{CF} + \mathrm{PI}$ and the $f(D)$ notation are ★★★; recognising constant against variable coefficients (8 Legendre/Cauchy occurrences need it) and homogeneous against non-homogeneous ★★; definitions and the linear-dependence test ★ (never asked; the Wronskian is used in class 011, variation of parameters).

3 · The differential operator D and $f(D)$

ESA Jan 2024 · 5 marks · UE23MA141A set 2, Q2.d · solved in W3.1

Solve $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$. (Here: reading $f(D)$; full solution in class 005.)

5 · The general solution: $y = \mathrm{CF} + \mathrm{PI}$

ESA Dec 2023 · 7 marks · UE22MA141A Q3.a (also May 2024 Q3.b, June 2024 Q2.a) · solved in W5.2

Solve $(D-2)^2 y = 8(e^{2x} + \sin 2x)$. (Here: the CF + PI structure.)

ESA Jul 2023 · 7 marks · UE22MA141A Q3.b · solved in W5.3

Solve $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$. (Here: the PI of a sum.)